Kalasalingam Academy of Research and Education

ExamPrep AI

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Team ID: RAG-TEAM-ALPHA

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Project Trainer / Guide

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1. Introduction

ExamPrep AI is an AI-powered study assistant designed to help students prepare for examinations using their own study materials. It allows users to upload PDF notes, ask questions and receive AI-generated explanations.

The system uses Retrieval-Augmented Generation (RAG) to retrieve relevant information from uploaded documents before generating answers through a Large Language Model.

It also provides practice tests, automatic scoring, explanations and a student dashboard.

2. Problem Statement & Objectives

Problem: Students spend considerable time searching through lengthy PDF notes to find relevant information. General AI tools may also provide answers that are not based on their actual study materials.

Objectives:

- To develop an AI-based study assistant.
- To upload and manage PDF study materials.
- To retrieve relevant information using TF-IDF and cosine similarity.
- To generate context-based answers using the Groq API.
- To provide AI-generated practice tests and automatic evaluation.
- To track student performance through a dashboard.

3. System Architecture (RAG Pipeline)

The ExamPrep AI system consists of two main phases: Document Ingestion and Query Processing.

3.1 Document Ingestion Phase

- User uploads a PDF document.
- Backend processes the uploaded document.
- Text is extracted and prepared for searching.
- TF-IDF converts the text into numerical vectors.
- Document information becomes available for retrieval.

3.2 Query Processing Generation Phase

- User enters a question.
- The question is converted into a TF-IDF vector.
- Cosine similarity compares the question with document content.
- Relevant passages are retrieved.
- Retrieved content and the question are sent to the Groq LLM.
- The generated answer is displayed to the user.

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4. Implementation Details & Tech Stack

Component	Technology Used	Purpose
Frontend	React and Vite	Developing the user interface
Programming Language	Python	Core backend logic

Backend Framework	Flask	Handling API requests and application logic
Retrieval Technique	TF-IDF	Converting text into numerical vectors
Large Language Model	openai/gpt-oss-20b (via Groq API)	Generating AI responses from retrieved context
Frontend Language	JavaScript (JSX)	Managing frontend functionality
Similarity Method	Cosine Similarity	Finding relevant document passages
Development Tool	Visual Studio Code	Project development
Deployment Platform	Vercel and Render	Hosting frontend and backend

4.1 Main Technologies

React and Vite are used to develop the frontend, while Flask handles backend operations. TF-IDF and cosine similarity identify relevant information from uploaded documents. The Groq API connects the application with the configured language model to generate responses using retrieved context.

5. Results and Testing

The application provides an integrated platform for document-based learning and examination preparation.

Functional Testing:

- PDF upload and document management.
- Question answering using uploaded study materials.
- Relevant passage retrieval.

- AI response generation through Groq.
- Practice-test generation and submission.
- Automatic score calculation.
- Dashboard and test-history display.

The system is designed to simplify document searching and provide an interactive learning experience. Actual accuracy and response-time measurements require separate testing.

6. Conclusion & Future Scope

Conclusion: ExamPrep AI combines document retrieval and generative AI to provide an interactive study platform. By using TF-IDF, cosine similarity and the Groq API, it helps students retrieve information from their notes and receive AI-generated explanations. Practice tests and performance tracking further support examination preparation.

Future Scope:

In the future, ExamPrep AI can be enhanced with user authentication, cloud database integration, improved semantic search, OCR support for scanned PDFs, personalized learning recommendations and advanced performance analytics to provide a more effective and personalized learning experience.